DP4c. Real-time sound radiation (acoustic transfer) for objects broken, carved or built during play: a ladder from libraries of the fracture process's own fragments, solved beforehand, through rules on the piece's own surface to exact solves, with a measurable "very close" line

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Classification (CPC): G10K 15/02 (synthesis of acoustic waves); A63F 13/54 (game output involving acoustic signals); A63F 13/57 (simulating properties, behaviour or motion of objects in the game world); G06F 30/20 (design optimisation, verification or simulation); G06F 30/23 (the same, using finite element or finite difference methods); H04S 7/30 (control circuits for electronic adaptation of the sound field).

Related disclosures in this series: DP4 (sound made from the matter: modal impact sound from shape and material); DP4b (matched solves: exact per-mode sound radiation, the top tier of this paper); DP5 (the physel, the shape primitive the pieces are made of).

Derived means computed exactly from the formulas given here, on spheres (Rayleigh's series, exact at any ka) or on two-dimensional rounded boxes and planks (an exact boundary solve by fundamental solutions [20]); the 2D figures are an analogue for 3D box-like pieces. "One possible choice (an example)" marks a detail filled in to make the method buildable, where others work too.

1. Field

Acoustics and real-time simulation: the sound radiated by the vibration of objects created while a game, a virtual-reality world or another simulation runs (broken, carved or built), played from the moment they first sound and refined while they play.

2. Problem and known methods

2.1 The problem

In modal sound synthesis an object rings as a sum of damped sine waves, one per vibration mode. What reaches a listener from each mode depends on its radiation, its strength and phase in each direction (its acoustic transfer). For a shape made in play it is not known in advance, and an exact solve takes far longer than the time available. A sound that answers one's own action (a hit, a grab, a break) should reach the ear within a few tens of milliseconds (one possible target (an example): 18 ms). Once the output path and the making of a break's pieces are paid, a few milliseconds are left for the first sound's model. A piece's own modes need an eigen solve, and its exact radiation a boundary or wave solve per

mode (DP4b): each takes far longer.

So nothing exact is ready for a break's first sound, nor for the knocks of the next second: pieces falling 0.2-1.8 m land after 0.2-0.6 s and settle in about 1-3 s. Pieces also turn fast (often 10-60 rad/s), and a break makes hundreds of them.

2.2 Known methods

Method	What it does	Published figures (the authors')
Precomputed acoustic transfer [1]	A boundary-element solve per mode, fitted by multipole sources	Objects known in advance
Fracture sound with precomputed soundbanks (Zheng and James 2010) [2]	Each fragment plays a precomputed ellipsoid's modes and transfer, chosen by the fragment's inertia, frequencies scaled	About 500 times faster; differences barely audible with every fragment proxied
Acceleration noise with precomputed soundbanks (Chadwick, Zheng and James 2012) [3]	Rigid-body acceleration noise from precomputed ellipsoid proxies	—
Interactive acoustic transfer approximation (Li, Fei and Zheng 2016) [4]	Transfer updated when shape or material changes; multipoles to order 18	11.6-28.9 s an update; a listening study of transfer errors
KleinPAT (Wang and James 2019) [5]	GPU time-domain precomputation of many modes at once	Hundred- to thousand-fold faster precomputation
NeuralSound (Jin et al. 2022) [6]	A learned predictor of modes and transfer	Under 1 s per new object
Radiation efficiency [7]; Kirchhoff and Rayleigh integrals [8, 9]; component mode synthesis [10]; shape derivatives [11]	Standard engineering and mathematical tools	—

2.3 What these leave open

1. **The right shapes.** Proxy banks [2, 3] give a first sound with no solve, but from a generic family (ellipsoids), not from the fracture process that made the piece; and no method we found tabulates exact radiation over the debris shapes the simulation itself makes, though most pieces of a break are such debris.
2. **What to do while better results arrive:** we found no rule for when a ringing mode may switch to better data without an audible jump, or how often.
3. **Which approximation to trust where:** borrowing a proxy's directivity is common; we found no comparison showing when it is worse than none.
4. **A measurable "close enough"**, per approximation, judged on pieces that turn as they fly. Learned predictors [6] carry no error bound for a new shape.

3. Summary

A new piece's first sound is a lookup, never a solve, and nothing waits on a solve afterwards. Each mode climbs a ladder: T0 plays from libraries solved beforehand (the exact rigid-motion radiation over the debris kind's whole family of shapes, and per fracture recipe a library of its own fragments, matched by recipe and inertia and scaled exactly in size, material wave speed and air); T1 applies rules on the piece's own surface once its modes exist; T2 solves exactly on spare CPU cores; T3 runs exact matched solves in the background, loudest first, minus modes a level bound shows inaudible. A mode moves up at most three times, at the piece's next hit or ramped over one audio block when its level changes by under 1 dB, and never down. "Very close" is a line: in cells of one auditory band by 20 ms, at least 95% within 1 dB of exact and none beyond 3 dB, decays within the smallest steps heard, and a blind listening score of 80 or more against the exact.

4. Detailed description

4.1 Conventions and inputs

- Time factor e^(−iωt); k = ω/c; air at 20 °C: c = 343 m/s, ρ0 = 1.2 kg/m³; c(T) = 331.3 + 0.606 T (T in °C).
- A piece has an outer surface Γ with outward normal n, mass m, principal moments of inertia $I_1 \geq I_2 \geq I_3$, a size L (its largest side) and a, the radius of the smallest sphere about its centre that encloses it.
- Mode n: angular frequency ω_n; mass-normalized shape φ_n ($\int \rho |\varphi_n|^2 \, dV = 1$); $u_n = \varphi_n \cdot n$ on Γ. For unit modal amplitude the surface velocity is $v_n = -i\omega_n u_n$ and the boundary data is $q = \partial p/\partial n = \rho_0 \omega_n^2 u_n$, as in DP4b.
- A mode's radiation efficiency is $\sigma = P / (\tfrac{1}{2} \rho_0 c \int_\Gamma |v_n|^2 \, dS)$, P its radiated power. At distance r in direction $\hat{d}$, $|p|^2 = \rho_0 c \, P \, D(\hat{d}) / (2\pi r^2)$, D the directivity (mean 1 over the sphere). Amplitudes are peak values.
- A **break** is a deterministic event: its point, energy, the material's **cut-set recipe** (how it breaks: radial and concentric cracks for glass, dice for tempered glass, cells stretched along the grain for wood, chunks for stone) and a seed. The same inputs make the same pieces on every machine.
- Pieces of about 2 mm-2.5 cm are the **debris kind**: a convex shape with three sizes $a_1 \geq a_2 \geq a_3$, simulated as rigid bodies. Finer grains play a granular voice (DP4).
- A **cell** is one auditory band (one equivalent rectangular bandwidth, ERB(f) = 24.7 (4.37 f/1000 + 1) Hz [12]) by one 20 ms window. The **reference** is an offline double-precision solver set (finite elements for the modes, exact radiation solves) that gives the answer every stand-in is judged against (DP4).

4.2 The ladder

Tier	Ready	What plays	Where
T0	At once: a lookup when the pieces are made	The debris table; the recipe entry, scaled, with its power factor and its smoothed pattern or power only	A physics worker and the audio thread
T1	When the piece's own modes land	Its own modes with the power rule and the facing-surface rule; carve updates; per-band statistics for large objects	A core the frame leaves, once a mode
T2	Soon after T1	Exact: the low band (rigid-body moments, low-ka terms); dense boundary solves to ka 5	Spare CPU cores
T3	In the background	Exact matched solves (DP4b), loudest first, minus what the level bound clears	A background GPU share

Each mode carries a small state for the mixer (one possible layout (an example)): tier (u8), moves made (u8), triage flag (u8), one spare byte, the index of a pending result (u32), and the level change toward the listener that result would make (f32): 12 bytes. Results are kept with their piece, keyed by a digest of its shape, for all its later sounds. A carved or rebuilt piece is a new shape and starts its own ladder. Pieces too small or too short-lived to sound again (debris, pieces about to merge into a rubble heap or vanish) are never solved: their T0 sound is all they play.

4.3 The move rule

```
on result R for mode m, of tier t:
    if t <= m.tier or m.moves == 3: drop R            # never down; at most three
moves
    else: m.pending = R;  m.delta = R's level change toward the listener (dB)
at each strike of the piece (its onset sample):
    for each mode with a pending result: install it; m.tier = t; m.moves += 1
at each audio block boundary (for example 10 ms blocks):
    for each ringing mode with a pending result and |m.delta| < 1 dB:
        ramp its gain and pattern linearly over this block; install; m.tier = t;
m.moves += 1
    # a pending result with |delta| >= 1 dB waits for the next strike
```

A change in the matter (temperature, wetness, a carve) is not a move: the sound changes because the thing changed. A mode that is never struck again decays with what it has.

4.4 The debris table: exact at once

Rigid motion does not depend on the material, so one table serves every material. The debris kind's shape is fixed up to its three sizes, so the table spans its whole family: no proxy error, only interpolation error.

- **Axes:** the aspect ratios $r_2 = a_2/a_1$ and $r_3 = a_3/a_1$ in [0.1, 1], log-spaced (100 shapes, 10 × 10), and about 30 ka points up to the top of hearing for the largest debris (ka ≈ 6.7 at 29.4 kHz for a 2.5 cm piece).
- **Contents:** for each shape, ka point and rigid motion (three translations and three rotations about the principal axes), the radiated field as complex multipoles to order 18 (361 complex f32 coefficients): 100 × 30 × 6 × 361 × 8 B ≈ 52 MB (derived).
- **Baked:** by offline dense boundary solves, shape by shape; at low ka, for an ellipsoidal debris kind (semi-axes $a_1 \geq a_2 \geq a_3$), by the closed form below.
- **Low-ka closed form** (after Lamb [13]): an ellipsoid accelerating along principal axis i with $a_i(t)$ radiates a dipole, $p(r, t) = \rho_0 V(1 + \alpha_i) \cos\theta_i \, \dot{a}_i(t - r/c) / (4\pi c r)$, with $V(1 + \alpha_i) = 2V/(2 - A_i)$, $A_i = (2/3)\, a_1 a_2 a_3 \, R_D(a_j^2, a_k^2, a_i^2)$ (Carlson's symmetric elliptic integral) and θ_i the angle from axis i. A sphere has $A_i = 2/3$ and $V(1 + \alpha) = 1.5\ V$.
- **Lookup:** interpolate in $(\ln r_2, \ln r_3)$ and in ka, then scale to the piece's own a_1 (§4.5). The interpolation error, on the ellipsoid's dipole factor interpolated in its logarithm, against the closed form at 400 random shapes: worst 0.071 dB linear and 0.002 dB cubic with 49 table shapes; 0.031 and under 0.001 dB with 100 (derived).
- **Playback:** each contact's acceleration pulse (DP4's contact model) drives the six rigid motions through the table's transfer toward the listener, in the piece's frame.

Why this covers so much. A compact solid's first elastic mode sits at $ka \approx \lambda_1 \, c_s/c_{air}$ at any size: about 13 for steel and granite, 14 glass, 10 concrete, 8 ice, 5 hardwood across the grain (derived). A stiff piece under about 5 cm therefore rings elastically only above about 29 kHz, the top of hearing: what is heard of it is its rigid motion, which the table gives exactly. By count, over 99% of a shattered brick's pieces are such debris (§6).

4.5 The recipe libraries

An entry. A library holds 1,000-4,000 fragments of one recipe at one size, each made by the recipe itself with its own seed and each mode solved exactly offline. Per entry, a header (one possible layout (an example)): recipe id and seed (2 × u32), size L_0 (f32), the moment ratios (2 × f32), volume (f32), mode count (u16), and 64 contact points in the entry's principal frame, normalized by L_0 (64 × 3 × f16). Per mode, 360 bytes:

Field	Type	Bytes
Dimensionless frequency $\Omega = \omega L_0/c_s$	f32	4
Loss factor of the recipe's material	f16	2
The two ka points, as factors of the nominal ka	2 × f16	4
Power factor σ at each ka point	2 × f16	4
Surface factor $s = \tfrac{1}{2} \rho_s L_0 \int_\Gamma \lvert u_n\rvert^2 \, dS$ (dimensionless)	f16	2
Low-band moments: monopole and dipole	4 × complex f16	16
Order-4 pattern at each ka point, normalized to unit power	2 × 25 × complex f16	200
Contact samples: $\varphi_n \cdot n \times \sqrt{(\rho_s L_0^3)}$ at the 64 points	64 × f16	128

That is 7-22 KB for a compact fragment with 20-60 modes and 11-108 KB for a shard or plank with 30-300, so with 1,000-4,000 entries **7-86 MB a compact recipe** (stone, concrete, tempered dice) and **11-430 MB a shard or plank recipe** (glass panes, wood, sheet). Eight recipes take about 0.06-0.7 GB if all are compact and up to about 3.5 GB if all are shards or planks; one possible choice (an example) where storage is tight, such as a standalone headset, is to cap shard and plank libraries at 1,000 entries

(11-108 MB each). Full-order multipoles would take about 80-2,880 MB a compact recipe.

The bake. Offline, in a fixed order with fixed seeds, so the same library comes out every time; re-run when the recipe or the solver changes. Each fragment's modes are solved by matched solves (DP4b). **Hero sets** (optional): an authored breakable object's own fragments from 8-32 sample breaks, baked the same way. The libraries ship with the program, not inside any world's content, and are identical on every machine; as pieces are deterministic too, every player hears the same T0 sound.

Finding an entry.

```
t0_entry(piece, recipe):
    (I1, I2, I3) = principal moments of the piece
    key = (ln(I2/I1), ln(I3/I1))                  # shape only: size is scaled
exactly below
    e = kd_tree[recipe].nearest(key)              # ties broken by entry index
    γ = (piece.volume / e.volume)^(1/3)
    return e, γ
```

Carrying it to the piece, exactly.

Change	Rule	Error
Size	Similar shapes scale exactly [2]: x → γx, ω → ω/γ, ka unchanged; mass-normalized mode shapes γ^(−3/2); multipole coefficients γ^(−5/2)	None
Material wave speed	ω = Ω c_s(T)/(γL0); exact at the same Poisson ratio (a recipe has one material; temperature changes E, and ν only a little)	None in pitch
Air	ρ0c exactly in the power; ka = ωa/c(T): air at −20 °C raises ka 7.5%, at +40 °C lowers it 3.5%	Below
Between the two ka points	σ and the pattern linearly in ka	Below
Damping	Material: the loss factor; radiation: σ_rad = σ ρ0c s/(ρ_s γL0) (from §4.6)	As σ

Pattern error at a shifted ka (spheres, ka 5-40, 95% of directions, derived): one stored point used as it is, 0.11-0.36 dB at a 3.5% shift, 0.38-1.10 dB at 10%, 1.4-2.9 dB at 23% (steel at 500 °C rings 23% lower). Two stored points bracketing the target, linear between them: 0.02-0.09 dB when they are 4% apart, 0.10-1.02 dB when 9% apart. One possible choice (an example): points at 0.97 and 1.03 of the nominal ka, which cover air at about 3-37 °C with the matter at 20 °C.

What carries, and what does not (derived):

- **Power carries:** a nearby shape's power is 0.00-0.04 dB off for compact 2D pieces; 1.3-1.7 dB (worst 2.6) for thin parts below coincidence.
- **Detailed directivity does not.** A surface offset δ between entry and piece scrambles a mode's level in each direction by about kδ, and 1 dB (12% in amplitude) needs kδ ≤ 0.12: offsets under 3.3 mm at 2 kHz, 0.44 mm at 15 kHz, 0.22 mm at 29.4 kHz. An inertia match may leave the shape 5-15% of the size off. And an entry's modes are another fragment's, so its pattern is another pattern. On spheres (ka 10-40), another piece's pattern with its power put right is 5.3-7.0 dB off at the median and 17-19 dB at 95% of directions for one still mode, against 3.5-3.9 and 9.6-10.9 dB for the right power with no directivity at all; for five modes turning 90°, 1.2-1.8 and 3.9-4.9 dB against 1.0-1.2 and 3.1-3.5. **So T0 never borrows a detailed pattern.** It plays the entry's order-4 pattern only where kδ_match ≤ 0.12 (δ_match the inertia mismatch times the size: one possible choice (an example)), and above that the power alone, spread evenly.
- **Pitch:** an entry's modes may be 1-5% off the piece's own, inside one auditory band (11-13% wide); the piece's own modes replace them at T1. A mode's strike strength is read at the contact sample nearest the hit.

4.6 The rules

The power rule. Where a mode's surface pattern outruns sound in air (its surface wavelength is longer than the air's: supersonic), σ ≈ 1, so

P = ½ ρ0c ∫_Γ $|v_n|^2$ dS, and the radiation damping (amplitude decay rate) σ_rad = P/ω_n^2 = ½ ρ0c ∫_Γ |

$u_n|^2$ dS,

a frequency-independent sum over the surface (derived). It holds for every compact solid's elastic modes (pattern order about ka ÷ (c_s/c_air)) and for thin parts above coincidence, $f_c = (c^2/2\pi)\sqrt{(\rho_s h/D)}$, $D = Eh^3/(12(1 - \nu^2))$: glass 4 mm 3.0 kHz, 6 mm 2.0 kHz; steel 1 mm 12.0 kHz, 5 mm 2.4 kHz; aluminium 1 mm 12.0 kHz; plywood 12 mm 1.4 kHz; hardwood 20 mm 0.7 kHz; concrete 50 mm 0.4 kHz (derived). One possible test (an example) for other modes: the surface wavenumber $k_s^2 = \int|\nabla_\Gamma v_n|^2$ dS / $\int|v_n|^2$ dS below k^2.

Case (derived)	Exact σ (the rule takes 0 dB)
Spheres, pattern order ka ÷ (c_s/c_air), c_s/c_air 4-9.5	−0.03 to +0.16 dB from ka 5; −0.97 dB at ka 2
Spheres where the pattern's order meets ka (the sphere's coincidence)	+1.1 dB (ka 5) to +5.9 dB (ka 80)
2D rounded boxes, ka 5-40	+0.02 to +0.50 dB
2D planks above coincidence, ka 2-20	0.0 to +0.8 dB
2D planks below coincidence	−10 to −25 dB: the rule fails
Low ka (a dipole: −18 dB at ka 0.5, −7 dB at ka 1)	The rule fails: the debris table and the low band (T2) cover it

The facing-surface rule. A mode's level toward the listener follows the piece's own surface that faces the listener (Figure 3). Each lit element ($n\cdot\hat{d} > 0$ and not hidden by the piece itself) radiates as if set in a rigid baffle (physical optics, the Rayleigh integral [8, 9]). In the far field $p \approx F(\hat{d})\, e^{ikr}/r$, with

$F(\hat{d}) = -(1/2\pi) \int_{lit(\hat{d})} e^{-ik\,\hat{d}\cdot y}\, q(y)\, dS(y)$.

On a curved, convex part the integral's stationary point y*, where $n = \hat{d}$, gives its local form, the **facing patch**:

$|p|\, r \approx \rho_0 c\, |v_n(y^*)| / \sqrt{K(y^*)}$, K the Gaussian curvature at y* (derived by stationary phase; exact on a sphere at high ka).

At T1 a PC evaluates F at about 1,000 directions and fits complex multipoles once a mode (with about 7,000 surface samples at λ/4 on a 10 cm piece), scaled to the power rule's power where that holds (one possible choice (an example)). A standalone headset evaluates the facing patch directly, about 50 flops a mode and path. On flat faces, where K = 0, one possible choice (an example) is each lit face's power spread over its half-space with a cosine weight, its error measured by the reference.

Case (derived)	Median / 95% of directions within 20 dB of the loudest
Spheres, local form, one still mode, ka 10-40	0.09-0.22 / 1.1-8.4 dB
Spheres, one mode turning 30° in a window, ka 20-40	0.09-0.11 / 0.87-0.99 dB
Spheres, three modes in the band, turning 30°	0.06-0.12 / 0.43-0.82 dB
2D rounded boxes, lit-element sum, still, ka 5-40	0.4-1.2 / 1.8-5.0 dB
2D rounded boxes, turning 30°	0.2-1.1 / 0.9-2.7 dB
2D planks above coincidence	0.3-2.6 / 2.2-11.7 dB (good from ka 20)
2D planks below coincidence	3.4-4.0 / 12.5-15 dB: the rule fails

Smoothed directivity and the speckle limit. At high ka a mode's level in one direction scatters about its mean like the intensity of a random wave field (exponentially, as in room acoustics [14]). A smooth stand-in (the right power, spread evenly) then errs by a gamma-distributed amount: with L independent looks in a cell (modes in the band × lobes swept by the turning piece), the error's standard deviation is $(10/\ln 10)\sqrt{\psi_1(L)}$ dB, ψ_1 the trigamma function. L = 1: 3.2 dB median, 13 dB at 95%; L = 5: 1.3 and 4.1 dB; L = 20: 0.7 and 1.9 dB (derived). A spread of 1 dB needs 20 looks, 0.5 dB 76, 0.24 dB 328. A turning fragment with 1-3 modes a band has L of about 3-15: 2-7 dB at 95% of cells. That is what T0 alone gives compact fragments, and why T1 follows.

Per-band statistics (dense modes). Where a band holds many modes the speckle limit is small. At T1 a band with at least 20 modes plays each mode at its own pitch and decay, with the power rule's power spread evenly; its error is the speckle limit with L = the band's count, taken from the piece's own mode

list. Examples (derived): a 50 cm steel shell with 2,000 modes to 29.4 kHz has 5.4 modes an ERB at 500 Hz (1.97 dB spread), 16 at 2 kHz (1.09), 60 at 8 kHz (0.56) and 149 at 20 kHz (0.36); a 50 cm granite block reaches 20 a band near 10 kHz (Weyl's law [18]); a 10 cm piece never does.

Carve updates. A carve changes the shape a little, so the last exact solve is updated, not thrown away:

1. Modes by first-order shape derivatives (Hadamard [11]). With w the boundary's outward normal displacement (negative where material is removed) and C the elasticity tensor: $\delta(\omega_n^2) = \int \delta\Gamma\, w\, [\varepsilon(\varphi_n):C:\varepsilon(\varphi_n) - \omega_n^2\rho|\varphi_n|^2]\, dS$, *and the shapes mix,* $\varphi_n' = \varphi_n + \Sigma(m\neq n)\, c_{nm}\, \varphi_m$, with $c_{nm} = \int_{\delta\Gamma} w\, [\varepsilon(\varphi_m):C:\varepsilon(\varphi_n) - \omega_n^2\rho\, \varphi_m\cdot\varphi_n]\, dS \,/\, (\omega_n^2 - \omega_m^2)$.
2. The unchanged surface: radiation is linear in the surface velocity, so the new mode's coefficients are $a_n + \Sigma\, c_{nm}\, a_m$, from the last exact solve.
3. The changed surface: add the facing-surface rule's field of the new patch and subtract that of the old one (the lit-element sum over the changed region only).
4. The low band re-solved (T2's coarse solve).

2D, a 3% notch (derived): the last solve alone is 0.05-0.26 dB off at the median and 0.21-3.1 dB at 95% of directions; with step 3, 0.04-0.17 and 0.32-1.25 dB (at ka 5 the 95% figure rose from 0.21 to 0.47 dB; at ka 10-40 it fell). An error indicator (the carved volume, the size of step 1's terms, step 3's level change) queues the shape for T3.

Component mode synthesis (built objects). An object built in play from parts with solved modes gets its modes from theirs (Craig and Bampton [10]): each part's fixed-interface modes plus constraint modes at its joints, joined through the joints' stiffness. A small eigenproblem gives the assembly's modes, exact to the kept mode sets and the joint model. Radiation: each part's solve where it stays exposed, the facing-surface rule on the assembly's exposed surface, and the low band on the union (T2).

4.7 Exact on spare cores (T2) and in the background (T3)

T2 uses spare CPU cores once the shape exists:

- **The low band** on a coarse surface (500 panels): one factored Laplace operator gives the rigid-body moments and the low-ka terms (ka < 0.5) of up to 100 modes. Rigid-body moments need only the shape, so they come first, before the modes.
- **Dense boundary solves**, one mode at its own frequency: a Burton-Miller equation [15] collocated on panels of λ/8 (at least 300), assembled at about 400 flops an entry and LU-factored ((8/3)N³ flops); a nearby mode reuses the factorization as its preconditioner (re-assembly and about 10 GMRES steps). Up to ka 5 that is 300-509 unknowns.

Piece (sphere-like, a = L/2)	2 cm	5 cm	10 cm	25 cm	50 cm
Low band (ka < 0.5) below	2.7 kHz	1.1 kHz	550 Hz	220 Hz	110 Hz
Dense solves (ka ≤ 5) below	27 kHz	11 kHz	5.5 kHz	2.2 kHz	1.1 kHz

T3 is DP4b's exact matched solves, at a background GPU share, never inside a frame's budget. The queue runs loudest first. One possible priority (an example): the mode's expected level above threshold at a typical listening distance, times its tier's error class, so thin parts below coincidence and large objects' middle bands, where no rule reaches 2 dB, go first.

Triage by a level bound. A mode whose highest possible level stays under the best ears' threshold never needs T3:

$P_{max} = \sigma_{peak}\, \rho_0 c\, E_{max} \int_\Gamma |u_n|^2\, dS$ (all of the hardest strike's energy E_{max} in this one mode; derived from §4.6),

$|p_{max}|^2 = \rho_0 c\, P_{max}\, (N + 1)^2 / (2\pi\, r_{min}^2)$ (directivity at most $(N + 1)^2$ for a field of order N, about ka plus a few),

and the mode is cleared if $20 \log_{10}(|p_{max}| / (\sqrt{2} \cdot 20\ \mu Pa))$ at its frequency stays below the threshold of the most sensitive listeners (one possible choice (an example): a free-field threshold curve lowered by a fixed margin). σ_{peak} is per kind: on spheres the peak over patterns is +1.1 to +5.9 dB (ka 5-80); elsewhere the reference measures it.

4.8 Scale, materials, air and water

Case	What changes
Materials	c_s/c_air from 3.8 (hardwood across the grain) to 9.9 (glass): the power rule holds for compact pieces from ka 5 (derived); soft materials bring elastic modes into T2's reach; wood's grain may need its own entries
Hot and cold	Air: ka by c(T), power by ρoc. Matter: pitch by √E(T) (steel at 500 °C rings at 0.77 times). Both through the two ka points; T1 recomputes its rules at the new pitch
Water	ka falls 4.3 times (1,480 against 343 m/s), so T2 reaches 4.3 times higher (a 10 cm piece's dense reach 5.5 → 23.5 kHz); heavy loading lowers pitch, to first order for compact pieces; thin parts may need a coupled solve; entries per medium
Rooms	Reflections add directions: more looks a cell, so smaller speckle errors

4.9 "Very close": the line, and how it is judged

Part	Definition
Cells	One ERB band × one 20 ms window, over the first 2 s after each strike; only cells within 40 dB of the loudest are judged
Level	At least 95% of cells within 1 dB of exact; none beyond 3 dB
Decay	Each audible mode's decay time within 24% for rings under about 1 s, 9.5% for longer (the smallest differences heard, as the method takes them)
Pitch	Once the piece's own modes play: within 0.083%, air loading included
Listening	Double-blind ABX [17] first, exact against stand-in; then MUSHRA [16] with the exact as hidden reference: 80 or more ("excellent") is very close

Why 1 dB. In Li, Fei and Zheng's listening study [4], a transfer error of 8% (about 0.7 dB) was chosen as the closer of two sounds 32.5% of the time, against 47.5% for an exact copy; an error of 19% (1.5 dB) only 5% of the time. 1 dB sits where a difference starts to be heard: slightly imperfect, very close.

How it is measured, per recipe and class, each tier reported apart with the share of play it covers:

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for each recipe: 50 or more pieces × 5 strike points × the reference's direction set
                 × {free field, two rooms}:
    simulate the strike and the piece's motion, its spin included
    for each listener path and audio sample: the listener's direction in the piece's
frame
    exact    = the reference's solve, evaluated along that path
    stand_in = the tier's result, evaluated along the same path
    E_x, E_s = energy in each cell (ERB band × 20 ms window), for both
    keep cells with E_x within 40 dB of the loudest;  e = 10 log10(E_s / E_x)
    pass if share(|e| <= 1 dB) >= 95% and max |e| <= 3 dB, with decays and pitch as
above
```

Turning pieces. At 10-60 rad/s a piece turns 11-69° in one window (derived), so the listener's direction sweeps an arc across the mode's lobes and the cell's energy averages them (Figure 4). Both answers are evaluated along the same simulated path, never at one fixed direction. The listening scenes, one possible set (an example): glass breaking, a stone chunk falling, a carved bar struck, a built shelf struck, a 50 cm slab struck. **A tier that misses** still plays (real time or nothing); its kind moves up the solve queue and the miss is reported. The same measurement checks each tier apart on exact 3D solves of recipe fragments, the debris table at high ka against dense solves, and carve updates against re-solves.

4.10 What runs where

Part	Where	Work	Standalone headset
To lookup	A physics worker	One k-d tree query a piece	Same
To playback, order 4	Audio thread	1,600 flops a mode-update (two ears, six paths)	Same
The facing patch	Audio thread	About 50 flops a mode and path	Same
T1 multipoles	A spare core	About 1,000 directions and one fit a mode,	Not run

Part	Where	Work	Standalone headset
		once	
T2	Spare cores	One factored Laplace operator a shape; (8/3)N³ flops for a first mode's factor, about 10 GMRES steps for a nearby one	Not run
T3	A background GPU share	DP4b's matched solves	Authored objects solved beforehand
Libraries	Disk	The debris table 52 MB; 7-86 MB a compact recipe, 11-430 MB a shard or plank recipe	Same, or shard and plank libraries capped at 1,000 entries

A standalone-headset profile (one possible choice (an example)): shapes made in play stay at T0 and the facing patch; authored objects play radiation solved beforehand.

5. Variants

The method may also:

- **Use other tiers:** more or fewer; other edges (any low-band edge below ka 1; dense solves to whatever ka the cores allow, or on a GPU); skipped tiers; T2 or T3 on another machine, a server or a cloud machine, shared with every player who hears the piece (a break's pieces are deterministic, so one solve serves every copy).
- **Move differently:** thresholds of 0.5-3 dB; ramps of one to several blocks, or cross-fades; more or fewer than three moves; moves at a zero crossing, when the ring falls below a level, or only at hits.
- **Build other libraries:** per material, recipe, size range, thickness or grain angle; entries at several sizes; a few hundred to tens of thousands of entries; hero sets; entries from measured objects; libraries shared across products.
- **Find entries differently:** other keys (moments, volume, surface area, hull or box ratios, curvature or spectral descriptors, the break's seed and cell index); k-d trees, ball trees, grids or hashes; several nearest entries with their powers averaged; a learned descriptor used only to find entries, the error still measured.
- **Store entries differently:** orders 0-10; f16, f32 or block-compressed values; patterns on a direction grid or as equivalent sources; one, three or more ka points, or a fit in ka; other contact-sample counts or a surface expansion.
- **Tabulate other families exactly:** any shape the simulation makes from few parameters (spheres, ellipsoids, superquadrics, boxes, cylinders, capsules, rods, plates, shells), for rigid motion or elastic modes, interpolated linearly, cubically or by splines in log axes.
- **Use other rules:** a radiation-efficiency curve per kind in place of σ = 1 (panel formulas as in [7]); the facing-surface rule with edge-diffraction terms or shadowing by other pieces; statistics over third octaves or Bark bands; carve updates to second order, or boundary-perturbation series; reduced bases over smooth shape families.
- **Triage and judge differently:** by masking, the listener's own hearing profile, distance and occlusion, or a device budget; a tighter (0.5 and 2 dB) or looser (2 and 6 dB) line; other cells (third octaves, gammatone filters, 5-50 ms windows); triangle tests or paired comparisons; lines per class, or on loudness models.
- **Run elsewhere:** CPUs with vector units, any maker's GPU through vendor-neutral interfaces, phones, standalone headsets, consoles, DSPs, FPGAs, cloud machines.
- **Serve other fields:** film and animation foley (breaking glass, debris and crashes heard at once while editing, refined in the background); simulation and training (engineering, safety drills, demolition); product acoustics (rattles, drop tests of packaging, a design change heard at once); robotics (impact and contact sounds for perception; digital twins); architecture (impact noise on floors, panels cut or assembled); instruments made in play; ultrasound and non-destructive testing; electromagnetics, where a resonator's radiation is looked up, estimated by physical optics and refined the same way.

6. Worked example

A shattered 2 kg brick (a power-law split of the mass with exponent 1.6 [19]): about 5 pieces above 3 cm (77% of the mass), about 780 of 2 mm-3 cm (22%), 0.9% finer; about 150 pieces above 5 mm.

1. The break makes its pieces. At once, the debris gets the debris table (exact rigid motion), the 5 large pieces get entries from the stone-chunk library, found by their moment ratios and carried over in size, wave speed and air (§4.5), and grit a granular voice: the crack and the first knocks play now.
2. When the large pieces' own modes arrive (T1), they play with the power and facing-surface rules; T2 solves their low bands and their modes up to ka 5.
3. In the background, kept pieces get matched solves (T3), loudest first, minus the modes the level bound clears; pieces that merge into a rubble heap are never solved.

For compact fragments the derived figures give 2-7 dB at 95% of cells at T0 (the speckle limit with 3-15 looks a cell, §4.6) and 0.9-2.7 dB once the facing rule plays (2D, turning 30°).

7. Figures



Figure 1. The ladder after a break (schematic). Time on a log scale. T0 plays at once, a lookup when the pieces are made; T1 arrives with the piece's own modes, T2 soon after on spare cores, T3 in the background. Shading marks when each result can be ready.



Figure 2. The first sound. Debris reads the debris table, exact for its whole family of shapes and every material. Larger pieces find the nearest of their recipe's own fragments by moment ratios and carry it over exactly.



Figure 3. Two rules on the piece's own surface. Left: a surface wave longer than sound's wavelength radiates fully (the power rule); a shorter one cancels. Right: the facing-surface rule sums the lit elements; on curved parts its local form reads the velocity at the point y* that faces the listener.

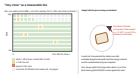


Figure 4. The very-close line, judged with turning pieces. Left: cells of one auditory band by 20 ms (schematic). Right: in one window the listener's direction sweeps an arc in the piece's frame; both answers are evaluated along it.

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